

PERTH OBSERVATORY, Australia

WMO: 956100

Lat: 32.007S

Lon: 116.136E

Elev: 385

StdP: 96.78

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 4.3	(c) 5.3	(d) -2.4	(e) 3.2	(f) 19.3	(g) 0.0	(h) 4.0	(i) 17.2	(j) 10.2	(k) 13.0	(l) 9.4	(m) 12.8	(n) 4.3	(o) 100	(p) 0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 13.9	(c) 34.9	(d) 18.5	(e) 33.1	(f) 18.4	(g) 31.4	(h) 18.1	(i) 21.2	(j) 28.8	(k) 20.3	(l) 28.3	(m) 19.5	(n) 27.6	(o) 3.6	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.5	23.4	18.0	13.5	22.6	16.9	12.6	21.6	63.4	28.5	59.9	28.3	57.1	27.8	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.6	7.9	2.7	38.5	0.9	1.6	2.1	39.6	1.5	40.6	1.0	41.5	0.4	42.6
			1.9	22.8	1.0	1.3	1.2	23.7	0.6	24.5	0.0	25.2	-0.7	26.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	22.6	22.6	20.9	17.7	14.4	11.9	10.9	11.4	12.4	14.9	18.2	20.7
	DBStd	5.13	3.29	3.18	3.48	3.04	2.40	1.93	1.86	2.03	2.49	3.00	3.47	3.74
	HDD10.0	31	0	0	0	0	1	5	12	8	5	1	0	0
	HDD18.3	1181	4	3	14	47	123	193	229	214	178	113	45	17
	CDD10.0	2413	389	353	337	232	138	62	41	53	78	153	245	333
	CDD18.3	522	135	123	93	29	2	0	0	0	1	8	41	91
	CDH23.3	5258	1464	1243	824	177	11	0	0	1	9	88	434	1007
(14) Wind	WSAvg	4.3	4.7	4.7	4.4	4.1	4.0	4.1	4.0	4.0	4.1	4.3	4.5	4.5
	PrecAvg	973	15	16	21	51	125	196	197	148	98	56	32	12
	PrecMax	1481	129	115	76	148	336	451	376	246	259	160	92	41
	PrecMin	629	0	0	0	0	21	30	90	17	12	1	2	0
	PrecStd	200	26	23	21	37	74	82	71	55	45	32	26	13
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	36.5	35.6	30.5	24.7	19.6	18.5	20.6	24.4	29.1	33.4	36.6
		MCWB	18.3	19.6	17.9	16.7	14.5	12.8	12.2	12.8	14.4	15.9	16.3	18.1
	2%	DB	34.6	33.9	32.5	27.6	22.3	18.1	16.7	18.3	21.4	25.6	30.5	33.9
		MCWB	18.7	19.0	18.3	16.3	14.2	12.4	12.1	12.4	13.5	15.3	16.5	17.9
	5%	DB	32.6	32.1	30.0	25.2	20.3	16.7	15.4	16.6	19.0	23.3	28.0	31.4
		MCWB	18.8	18.4	17.6	16.0	13.7	12.2	11.8	12.2	12.9	14.9	16.1	17.8
	10%	DB	30.5	30.2	27.9	23.1	18.6	15.5	14.2	15.2	17.0	21.0	25.7	28.9
		MCWB	18.4	18.2	17.4	15.5	13.6	12.1	11.5	11.8	12.3	14.2	15.7	17.2
	0.4%	WB	23.0	22.5	21.1	19.4	17.6	15.4	15.0	15.4	16.1	17.9	19.3	20.7
		MCDB	30.0	29.8	27.8	24.0	20.5	16.9	16.0	16.9	21.0	25.0	27.3	30.0
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.7	21.1	20.0	18.4	16.6	14.5	13.7	14.2	14.9	16.5	18.2	19.7
		MCDB	28.5	29.3	26.9	23.1	18.7	15.9	14.8	16.0	18.2	23.3	26.4	29.4
	5%	WB	20.4	20.1	19.2	17.5	15.6	13.7	12.8	13.2	14.0	15.6	17.3	18.8
		MCDB	28.2	28.5	26.3	22.2	18.0	15.1	14.1	15.2	17.1	21.2	24.8	28.4
	10%	WB	19.4	19.1	18.3	16.6	14.8	12.9	12.0	12.4	13.1	14.7	16.4	17.9
		MCDB	27.6	27.6	25.3	21.1	17.3	14.4	13.5	14.4	16.0	19.4	23.2	26.7
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	13.5	12.2	9.9	8.3	7.2	6.9	7.5	8.6	10.4	12.0	13.3
		MCDBR	16.8	16.0	15.1	12.7	11.0	8.9	8.3	9.7	12.2	14.5	16.2	16.8
		MCWBR	5.9	5.5	5.2	5.3	5.4	5.2	5.1	5.5	6.0	6.5	6.1	6.0
	5% WB	MCDBR	13.1	13.2	11.9	9.9	7.4	6.3	6.1	7.3	9.5	12.0	13.3	14.2
		MCWBR	5.7	5.6	4.9	5.0	4.7	4.8	4.8	5.1	5.4	5.9	5.9	5.9
(40) Clear-Sky Solar Irradiance	taub		0.323	0.323	0.309	0.301	0.293	0.286	0.286	0.291	0.311	0.311	0.313	0.312
	taud		2.596	2.612	2.645	2.653	2.647	2.650	2.647	2.626	2.541	2.563	2.583	2.608
	Ebn at Noon		1016	997	978	935	894	877	894	933	960	998	1020	1031
	Edh at Noon		104	99	90	80	71	67	70	80	98	103	105	103
(44) All-Sky Solar Radiation	RadAvg		8.18	7.36	6.08	4.43	3.31	2.71	2.87	3.71	4.94	6.45	7.70	8.37
	RadStd		0.40	0.27	0.30	0.27	0.22	0.18	0.18	0.19	0.28	0.30	0.35	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page